## CPSC 233 Assignments 04 Due Date: Tuesday, Aug 01th by 11:59 PM

# Topic: Reading and Writing from Files, Exception Handling

## Collaboration

Discussing the assignment requirements with others is a reasonable thing to do, and an excellent way to learn. However, the work you hand-in must ultimately **be your work**. This is essential for you to benefit from the learning experience, and for the instructors and TAs to grade you fairly. Handing in work that is not your original work, but is represented as such, is plagiarism and academic misconduct. Penalties for academic misconduct are outlined in the university calendar.

Here are some tips to avoid plagiarism in your programming assignments.

1. Cite all sources of code that you hand-in that are not your original work. You can put the citation into comments in your program. For example, if you find and use code found on a web site, include a comment that says, for example:
   `# the following code is from https://www.quackit.com/python/tutorial/python_hello_world.cfm.`
   Use the complete URL so that the marker can check the source.
2. Citing sources avoids accusations of plagiarism and penalties for academic misconduct. However, you may still get a low grade if you submit code that is not primarily developed by yourself.
3. Discuss and share ideas with other programmers as much as you like, but make sure that when you write your code that it is your own. A good rule of thumb is to wait 20 minutes after talking with somebody before writing your code. If you find yourself exchanging code by electronic means, writing code while sitting and discussing with a fellow student, typing what you see on another person's console, then you can be sure that "your" code is not substantially your own, and your sources must then be cited to avoid plagiarism.
4. **Collaborative coding is strictly prohibited.** Your assignment submission must be strictly your code. Discussing anything beyond assignment requirements and ideas is a strictly forbidden form of collaboration. This includes sharing code, discussing code itself, or modeling code after another student's algorithm. **You cannot use (even with citation) another student's code.**
5. We will be looking for plagiarism in all code submissions, possibly using automated software designed for the task. For example, see Measures of Software Similarity (MOSS - https://theory.stanford.edu/~aiken/moss/).

Remember, if you are having trouble with an assignment, it is always better to go to your TA and/or instructor to get help, than it is to plagiarize.

## Requirements:

**Question 1.** Write a program that will record the purchases made at a store. For each purchase, read from the keyboard an item's name, its cost price and the numbers bought. Compute the cost of the purchase. And write all this data to a text file. Also, display this information and the current total cost on the screen. After all items have been entered, write the total cost to both the screen and the file. Since we want to remember all purchases made, you should append new data to the end of the file.

**Submission:** Run this program 2 – 3 times. Store the record of the purchases in the text file. Submit a java file and a text file to the D2L.

**Rubric:**

Submission of text file (1 marks)

Submission of java file (1 marks)

Functionalities (6)

**Question 2:**

Suppose you are given a text file that contains the names of people. Every name in the file consists of first name and last name. Unfortunately, the programmer who created the file of names had a strange sense of humor and did not guarantee that each name was on a single line of the file. Read the file of names and write them to a new text file, one name per line. For example, if the input file contains

Bon Jones Fred

Charles Ed

Marston

Jeff

Williams

The output file should be

Bon Jones

Fred Charles

Ed Marston

Jeff Williams

**Submission:** Submit a java file and input text file to the D2L.

**Rubric:**

Submission of input text file (1 marks)

Submission of java file (1 marks)

Correct output file produced (6)

For each of the question, you should handle exception. For example, if files not does exist, your program should generate an exception. (4 marks)

**Credit:**

Some of the questions have been adapted from the course text book: Java An introduction to Problem Solving and Programming by Walter Savitch